Exercise

Q. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represent a pair of parallel straight lines, then prove that $h^2 = ab$ and $bg^2 - af^2 = 0$.

Solution: Given $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b}$$

Since pair of straight lines are parallel, we have $\theta = 0$ then $h^2 = ab$
(Expand the calculations yourself.)

We know that the condition of a pair of straight line is:

$$\Delta = 0$$
$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$
$$\Rightarrow h^2c + 2fg\sqrt{ab} - af^2 - bg^2 - ch^2 = 0$$
$$\Rightarrow -bg^2 + 2fg\sqrt{a}\sqrt{b} - af^2 = 0$$
$$\Rightarrow \left(\sqrt{b}\right)^2 g^2 - 2\sqrt{b}gf\sqrt{a} + (\sqrt{a})^2 f^2 = 0$$
$$\Rightarrow (\sqrt{b}g)^2 - 2\sqrt{b}gf\sqrt{a} + (\sqrt{a}f)^2 = 0$$
$$\Rightarrow (\sqrt{b}g - \sqrt{a}f)^2 = 0$$
$$\Rightarrow \sqrt{b}g - \sqrt{a}f = 0$$
$$\Rightarrow \sqrt{b}g = \sqrt{a}f$$
$$\Rightarrow (\sqrt{b}g)^2 = (\sqrt{a}f)^2$$
$$\Rightarrow bg^2 = af^2$$
$$\therefore bg^2 - af^2 = 0$$

Reduce the equation in standard form

Q. Reduce the equation in standard form $9x^2 - 6xy + y^2 - 14x - 2y + 12 = 0$. Hence identify it.

Solution: Given that,

$$9x^2 - 6xy + y^2 - 14x - 2y + 12 = 0$$

Now compare with the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, we get

$a = 9, h = -3, b = 1, g = -7, f = -1$ and $c = 12$

We have $\Delta =$ *"Find the value yourself"* $-400 \neq 0$

and $ab - h^2 = 0$. So given equation represent Parabola.

Now,

$9x^2 - 6xy + y^2 - 14x - 2y + 12 = 0$

$\Rightarrow 9x^2 - 6xy + y^2 = 14x + 2y - 12$

$\Rightarrow (3x - y)^2 = 14x + 2y - 12$

To choose coordinates along perpendicular directions, add a constant (k) inside the square.

$\Rightarrow (3x - y + k)^2 = 14x + 2y - 12 + k^2 + 6xk - 2ky$

$\Rightarrow (3x - y + k)^2 = x(14 + 6k) + y(2 - 2k) + k^2 - 12$.....(1)

Reduce the equation in standard form – Solution Continues

Let $3x - y + k = 0$ and $x(14 + 6k) + y(2 - 2k) + k^2 - 12 = 0$ are mutually perpendicular each other-

$3.(14 + 6k) + (-1).(2k - 2) = 0$

$\therefore k = -2$

From equation (1) we get

$$(3x - y - 2)^2 = x(14 + 6.-2) + y(2 - 2.-2) + (-2)^2 - 12$$
$$\Rightarrow (3x - y - 2)^2 = 2x + 6y - 8$$
$$\Rightarrow \left(\frac{3x - y - 2}{\sqrt{3^2 + 1^2}}\right)^2 .10 = \frac{2x + 6y - 8}{\sqrt{2^2 + 6^2}} .\sqrt{40}$$

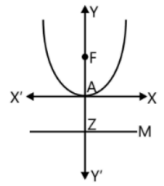
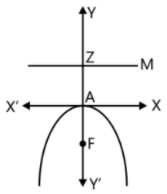
Let $X = \frac{3x-y-2}{\sqrt{10}}$ and $Y = \frac{2x+6y-8}{\sqrt{40}}$

$X^2 = Y.\sqrt{40}$

$$\Rightarrow X^2 = \frac{\sqrt{40}}{10} .Y$$
$$\Rightarrow X^2 = 4.\frac{\sqrt{40}}{40} .Y$$

$\Rightarrow X^2 = 4.\frac{1}{\sqrt{40}} .Y$ Here $a = \frac{1}{\sqrt{40}}$

Supporting material

Upward parabola: The equation of this type of parabola is of the form $x^2 = 4ay$, $a > 0$. See Fig. 10.9	 $x^2 = 4ay$ **Figure 10.9**
Downward parabola: The equation of this type of parabola is of the form $x^2 = -4ay$, $a > 0$. See Fig. 10.10	 $x^2 = -4ay$ **Figure 10.10**

Supporting material

Equation of the parabola Properties	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Vertex (Co-ordinates)	(0, 0)	(0, 0)	(0, 0)	(0, 0)
Focus (Co-ordinates)	(a, 0)	(–a, 0)	(0, a)	(0, –a)
Latus rectum (length)	4a	4a	4a	4a
Axis (Equation)	y = 0	y = 0	x = 0	x = 0
Directrix (Equation)	x = –a	x = a	y = –a	y = a
Symmetry (about)	x-axis	x-axis	y-axis	y-axis

Thanks for your attention